

Davide Avagnano

Software Engineer · Full-Stack — TypeScript · Node.js · PostgreSQL · Applied AI

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Profile

Full-stack Software Engineer with a **strong backend and architecture focus** in the **TypeScript / Node.js / React** ecosystem on PostgreSQL. I plan and model before writing, apply design patterns, and treat living documentation and planned refactoring as part of "done". Comfortable in UNIX-like environments (Linux, Bash, Docker), with hands-on experience in **applied AI in production** and SecOps. Engineer by training and self-taught developer: I recently designed and built an entire SaaS **end-to-end**, from architecture to production.

Experience

Software Engineer — Freelance / Consultant

2026 – Present

Baaarber — multi-tenant SaaS platform, end-to-end product ownership for a client

- Designed and built **almost single-handedly** a multi-tenant SaaS (**Next.js 16 App Router/RSC, React 19, tRPC, Drizzle ORM, PostgreSQL, Redis**): ~57k LOC TypeScript, 31 tables, ~145 type-safe procedures.
- Built an **applied AI layer** with cost governance: a streaming conversational assistant (Vercel AI SDK) and automated translation across 5 languages, with usage logs and per-model spend caps.
- Granular authorization (RBAC + plan gating) and **PII at-rest encryption** (AES-256-GCM) with boundaries enforced in CI; multi-channel notifications (email + push) with an outbox pattern and idempotent workers (BullMQ/PM2).
- Living documentation** approach (50+ technical and functional docs) and planned refactoring; deployed on Linux (Nginx, CI/CD via GitHub Actions).

Software Engineer — Freelance / Consultant

2025 – 2026

High-volume production platforms (ixily, Scalability, Hypefill) and SecOps tooling

- Development and **analysis of complex production code** on marketing-intelligence platforms (Node.js/TypeScript, PostgreSQL, Redis, Elasticsearch): config-driven ETL, cross-platform data reconciliation and distributed workers (15+ PM2 processes).
- Built **20+ external API integrations** (Meta Graph, Shopify GraphQL, TikTok, Taboola, GA4, call tracking) via reusable connectors.
- Developed **analytics dashboards** (React, Recharts) with multi-level RBAC and **AI-in-production** features (RAG assistants, content moderation).
- Contributed to a **SecOps automation tool** (TypeScript + Bash): Suricata IDS + Elasticsearch + UFW firewall, remote rule lifecycle via SSH, ELK stack.
- Collaboration in **small teams** (3–5 developers) with CTO, design and marketing, in requirements-gathering calls; supported colleagues on technical problems.

Software development — Self-taught transition

2021 – 2024

Career switch from engineering to software: React/Next.js web apps on public APIs, algorithms and data structures, data analysis in Python (pandas, NumPy).

Skills

Languages: TypeScript (advanced), JavaScript, SQL, Bash; Python (data analysis)

Frontend: React (17→19), Next.js (App Router / RSC), Tailwind CSS, shadcn/ui, TanStack Query, Zustand, i18n

Backend, architecture & security: Node.js, Express, tRPC (type-safe APIs), Zod, REST; authentication (better-auth/NextAuth, OAuth, JWT); event-driven design and distributed workers (BullMQ, PostgreSQL LISTEN/NOTIFY); granular RBAC, PII encryption (AES-256-GCM), domain modeling

Data & persistence: PostgreSQL (advanced), MySQL, Redis, Elasticsearch, Drizzle ORM; config-driven ETL and cross-platform data reconciliation; pandas, NumPy

AI / LLM: OpenAI API, Vercel AI SDK (streaming), RAG (hybrid BM25 + kNN), applied LLMs with cost governance

DevOps & UNIX: Linux (Ubuntu), Bash, Docker, Nginx, PM2, CI/CD (GitHub Actions), Git; SecOps (Suricata IDS, ELK, UFW, SSH)

Education

BSc in Aerospace Engineering — University of Naples Federico II

2016 – 2020 · 105/110

Languages

Italian (native) · English **B2** — professional working proficiency; daily reading and writing on codebases and technical documentation.